Open problems:
Unphysicality of Hilbert spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Further problems.** The list of ways in which QSTO:H fails the base conditions is open-ended: "If the reader is aware of other problems that should be added, do contact us." (Brief, Sec. 1, footnote 1.)

2. **A physically adequate model.** The brief concludes that "the solution is not to find workarounds for these issues, but to fix the mathematical model to better reflect the physical requirements". A replacement for QSTO:H that satisfies [B]COBS, [B]WDQ and [B]STDIST is not given here; one candidate is the brief "Physical vector spaces for quantum mechanics". (Brief, Sec. 6.)

3. **Schwartz spaces.** "This third failure is not related to the problems of continuity, as Schwartz spaces present the same problem." Presumably this means that $\mathcal{S}(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^m)$ are isomorphic as topological vector spaces, as both are isomorphic to the space s of rapidly decreasing sequences through the Hermite functions. This is not shown, and it is not said which structure the isomorphism has to carry. (Brief, Sec. 4, proof of Theorem 5, last paragraph.)

4. **Classical cutoffs.** For [B]COBS and [B]WDQ the brief says that in classical mechanics "one can, in principle, put a cutoff on all variables and work only with joint probability distributions in that compact region", which "bound all moments", while "Quantum mechanics does not allow that". Neither half is made precise. For the quantum half, one reading is that a nonzero wave function and its Fourier transform cannot both have compact support. (Brief, Sec. 4, proof of Theorem 5.)

From the formalization

1. **The position operators.** The unitary U between the two frames, its inverse, and the common domain $\{\psi : x\psi \in L^2\}$ of X and Y are formalized, and $\psi \in \mathcal{D}(X)$, $U\psi = \phi$, $U\psi \notin \mathcal{D}(Y)$ are checked. The position operators themselves, their self-adjointness, and the operator UXU^{-1} of multiplication by $g(y)$ with domain $U\,\mathcal{D}(X)$ are not formalized. (Brief, Sec. 2, Proposition 3; Log, "Differences".)

2. **Point-wise values.** That $\psi(x)$, and so $|\psi(x)|^2$, is not defined at each point of a vector of L^2 is not formalized. (Brief, Sec. 5; Log, "Differences".)

3. **Commutator relations.** That $[X, P]/(i\hbar) = I$ cannot hold on the whole of $\mathcal{H}$ and must be restricted to a domain is not formalized. (Brief, Sec. 5; Log, "Differences".)

4. **The Schrödinger equation with a potential.** The remark on the domain of the Schrödinger equation is checked for the oscillator in the number basis, $H = N$ on $\ell^2(\mathbb{N})$, and in position space for the free particle, $H_0 = P^2/2$ on $L^2(\mathbb{R})$, with $\mathcal{D}(P) = H^1(\mathbb{R})$. For $H = P^2/2m + V(x)$ with a potential, the domain and self-adjointness of H and its evolution e^{-itH} are not formalized. They need results such as the Kato–Rellich theorem and Stone's theorem, which Mathlib does not have; the oscillator in position space would also need the Hermite functions as a basis of $L^2(\mathbb{R})$. (Brief, Sec. 5; Log, "Differences".)

The blocker of the previous version of the brief, that the conditions and the inconsistency were informal, is resolved. The revised brief adds a minimum form to each condition (Sec. 3) and reads the inconsistency as the failure of a condition for one particular system (Sec. 4). The previous open problem on the expectations of a general unbounded observable is no longer raised: the [B]COBS part now rests on the number operator alone (Proposition 2). The formalization gap "Theorem 5 itself" is closed: Theorem 5 is stated and proved in Lean as `theorem5`, in the readings of the next section. No open problem now blocks verification.

Precise readings

The revised brief confirms the readings proposed for its previous version, with the changes below. They are the readings in which Theorem 5 is stated in Lean (`theorem5`).

Inconsistency. "QSTO:H is inconsistent with C" means: there is a quantum system to which QSTO:H applies whose QSTO:H model fails condition C. The systems are the one-dimensional harmonic oscillator for [B]COBS, a system with one spatial degree of freedom for [B]WDQ, and two systems with different numbers of continuous degrees of freedom for [B]STDIST. Under this reading finite-dimensional Hilbert spaces do not need to be excluded.

[B]COBS.

- *States* are the nonzero vectors of $\mathcal{H} = \ell^2(\mathbb{N}_0)$, with the norm topology; this is equivalent to the quotient topology on $P(\mathcal{H})$, since everything below is invariant under $\psi \mapsto c\psi$.
- The observable is the number operator N with domain $\mathcal{D}(N)$ of Proposition 2.
- The *expectation* of N is $E_N(x) = \langle x|Nx\rangle / \|x\|^2$, defined for $x \in \mathcal{D}(N) \setminus \{0\}$.

B COBS, minimum form: E_N is continuous on $\mathcal{D}(N) \setminus \{0\}$ with the subspace topology.

- Claim to be proved: E_N is not continuous at e_0 within $\mathcal{D}(N) \setminus \{0\}$ (Proposition 2; `expN_not_continuousWithinAt` and `not_continuousOn_expN`, checked). That N is self-adjoint on $\mathcal{D}(N)$ and that $E_N(x) = \langle x|Nx\rangle / \|x\|^2$ there are checked too (`isSelfAdjoint_numberOp`, `coe_domain_numberOp`, `expN_eq_inner`).
- Not claimed by the brief, but true: for every densely defined, symmetric, unbounded O, $\psi \mapsto \langle O\rangle_\psi$ is discontinuous at every point of $\mathcal{D}(O) \setminus \{0\}$.

[B]WDQ.

- A *change of frame* is a smooth increasing bijection $y = f(x)$ of $\mathbb{R}$. It acts on states by the unitary $U\colon L^2(\mathbb{R}, dx) \to L^2(\mathbb{R}, dy)$, $(U\psi)(y) = \psi(g(y))\sqrt{g'(y)}$ with $g = f^{-1}$ (Proposition 3).
- Each observer measures position with the multiplication operator by their own coordinate, on its maximal domain: X on $L^2(\mathbb{R}, dx)$ and Y on $L^2(\mathbb{R}, dy)$. The brief now says this, and notes that UXU^{-1} is the different operator of multiplication by $g(y)$, with domain $U\,\mathcal{D}(X)$.
- A normalized state has *finite variance* for an observable O if it lies in $\mathcal{D}(O)$, that is, if $\|O\psi\|^2 < \infty$ (Sec. 4).

B WDQ, minimum form: for every change of frame and every state ψ, $\psi \in \mathcal{D}(X)$ implies $U\psi \in \mathcal{D}(Y)$.

- Claim to be proved: for $f(x) = \tan(\frac{\pi}{2}\operatorname{erf}(x))$ and $\psi = (e^{-x^2}/\sqrt{\pi})^{1/2}$, $\psi \in \mathcal{D}(X)$ but $U\psi = (1/(\pi(1 + y^2)))^{1/2} \notin \mathcal{D}(Y)$ (Proposition 3; `gaussState_mem_positionDomain`, `frameUnitary_gaussState`, `frameUnitary_gaussState_not_mem` and `not_forall_frameUnitary_mem_positionDomain`, checked).

[B]STDIST.

- The *systems* are those with $n \geq 1$ continuous degrees of freedom (Proposition 4 is stated for positive integers).
- The *mathematical representation* of a system is $\mathcal{H}_n = L^2(\mathbb{R}^n)$ with what QSTO:H builds from it: $P(\mathcal{H}_n)$, $D(\mathcal{H}_n)$, the self-adjoint operators, the Born rule and the von Neumann entropy. The brief now lists exactly these, so the positions $X_1, \ldots, X_n$ and momenta $P_1, \ldots, P_n$ are not part of the representation.
- Two systems have the *same* representation if a unitary $\mathcal{H}_n \to \mathcal{H}_m$ carries one onto the other.

B STDIST, minimum form: for $n \neq m$ there is no unitary $\mathcal{H}_n \to \mathcal{H}_m$.

- Claim to be proved: for all $n, m \geq 1$ there is one (Proposition 4; `L2_euclidean_iso`), and it carries the QSTO:H structure (`carriesQSTO`, `exists_carriesQSTO`). Both are checked.